AKS Node Pools Cheatsheet for Cost and Performance

Mixing VM sizes in Azure Kubernetes Service prevents app crashes and lowers your cloud bill. By matching specific workloads to specialized node pools, you pay only for compute power you actually need.

What you'll learn:

1. Separate System and User Node Pools
2. Slash Costs with Spot Node Pools
3. Direct Pods using Node Selectors
4. Enable Cluster Autoscaler per Pool

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Separate System and User Node Pools

Keep Kubernetes background services isolated from your application code. System node pools host CoreDNS, metrics-server, and cluster agents. User node pools run your microservices. Isolating them ensures an app memory leak won't crash your entire Kubernetes cluster control plane.

```
$az aks nodepool add --resource-group myRG --cluster-name myAKSCluster --name apppool --mode User --n
```

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Slash Costs with Spot Node Pools

Azure offers unused compute capacity at up to 90% discount through Spot VMs. Azure can evict these nodes with short notice, making them ideal for stateless background jobs, testing environments, and batch processing. Avoid using Spot pools for persistent databases.

```
$az aks nodepool add --resource-group myRG --cluster-name myAKSCluster --name spotpool --priority Spo
```

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Direct Pods using Node Selectors

After creating custom node pools (like memory-optimized E-series VMs), you must tell Kubernetes which pods should land on them. Add a simple nodeSelector block matching your pool's agentpool label inside your Deployment manifest.

```
spec:
  containers:
  - name: redis-cache
    image: redis:alpine
  nodeSelector:
    agentpool: mempool
```

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Enable Cluster Autoscaler per Pool

Stop paying for idle VM instances during low-traffic hours. The Cluster Autoscaler monitors pending pods and dynamically provisions or deletes VMs in a specific pool based on your min and max boundaries.

```
$az aks nodepool update --resource-group myRG --cluster-name myAKSCluster --name apppool --enable-clu
```

See Key Takeaways on next page >> linkedin.com/in/mramanrajsingh

Key Takeaways

+ Use smaller general-purpose VMs (like Standard_DS2_v2) for dedicated System node pools.
+ Reserve Spot node pools strictly for fault-tolerant and stateless workloads.
+ Configure nodeSelector or nodeAffinity to steer heavy workloads to dedicated pools.
+ Always define min-count and max-count when enabling the cluster autoscaler.
+ Mix CPU-optimized (F-series) and Memory-optimized (E-series) pools based on application needs.

Watch Out For

! Running heavy application workloads inside the System node pool.
! Deploying stateful production databases on Spot node pools.
! Forgetting to set pod resource requests, causing the cluster autoscaler to fail to trigger.
! Omitting node selectors, leading to heavy pods landing randomly on weak system nodes.

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